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3.8 Nuclear Physics

1. Radioactive Radiation

Radioactive decay: The spontaneous and random emission of radiation from an unstable nucleus.

- **Spontaneous:** The decay is not triggered by external conditions (temperature, pressure, chemical state, etc. have no effect). It happens because the nucleus is in an unstable energy state.
- **Random:** It is impossible to predict which particular nucleus will decay next, or exactly when any given nucleus will decay. We can only describe the probability of decay using statistical methods.

Types of Radiation

Property	Alpha (α)	Beta-minus (β^-)	Beta-plus (β^+)	Gamma (γ)
Nature	Helium nucleus (^4_2He)	High-speed electron ($^0_{-1}e$)	Positron ($^0_{+1}e$)	High-energy EM photon
Charge	+2e	−1e	+1e	0
Mass	4 u (heaviest)	~1/1836 u	~1/1836 u	0 (massless)
Ionising ability	Strongest (most ionising)	Moderate	Moderate	Weakest (least ionising)
Penetrating power	Weakest (stopped by paper or a few cm of air)	Moderate (stopped by ~3 mm aluminium)	Moderate (annihilates with electron)	Strongest (reduced by thick lead or concrete, never fully stopped)

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Deflection in E/B fields	Deflected (toward − plate; less curved due to large mass)	Deflected (toward + plate; more curved)	Deflected (toward − plate; more curved)	Not deflected
Speed	~5% of c (discrete energies)	Up to ~99% of c (continuous spectrum)	Up to ~99% of c (continuous spectrum)	c (speed of light)

Nuclear Equations

Alpha decay: The nucleus emits a helium nucleus (2 protons + 2 neutrons).

$$^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}\alpha$$

Beta-minus (β^-) decay: A neutron transforms into a proton inside the nucleus, emitting an electron and an electron antineutrino.

$$^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}\beta + \bar{\nu}_e$$

At the quark level: a down quark changes to an up quark ($d \rightarrow u + e^- + \bar{\nu}_e$), mediated by the W^- boson.

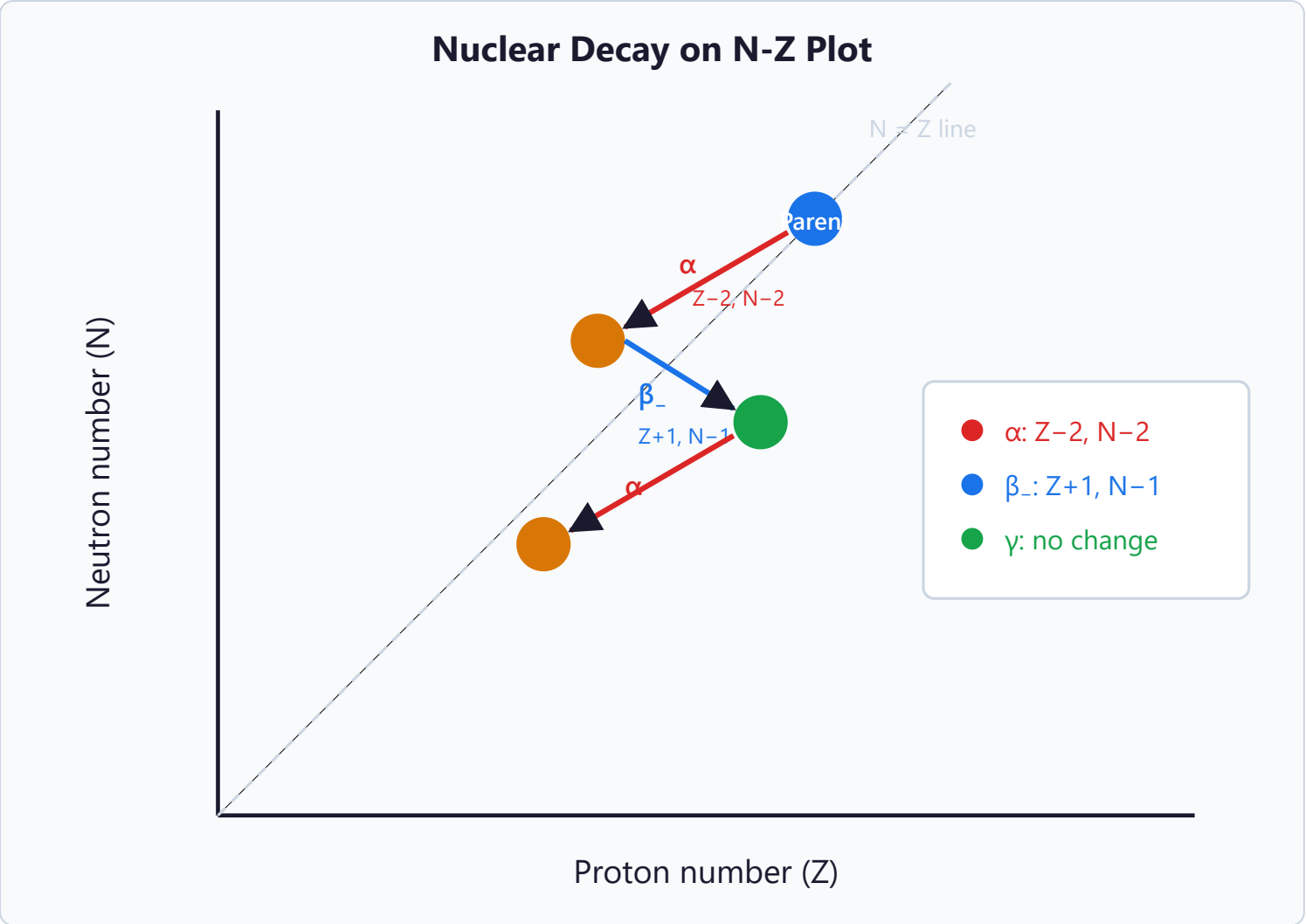
Beta-plus (β^+) decay: A proton transforms into a neutron, emitting a positron and an electron neutrino.

$$^{A}_{Z}X \rightarrow {}^{A}_{Z-1}Y + {}^{0}_{+1}\beta + \nu_e$$

Gamma emission: No change in A or Z. The nucleus loses energy by emitting a gamma photon. Often follows alpha or beta decay when the daughter nucleus is left in an excited state.

Conservation rules: In all nuclear equations, the following are conserved:

- Mass number (A) — total number of nucleons
- Charge number (Z) — total proton number / charge
- Lepton number
- Baryon number
- Energy and momentum



Exam Tip: The existence of the neutrino was proposed by Pauli (1930) to explain the continuous energy spectrum of beta particles. If only two particles were produced (daughter nucleus + electron), conservation of energy and momentum would require the electron to have a fixed, discrete kinetic energy. The observed continuous spectrum (from zero up to a maximum) implies a third particle (the antineutrino) carries away the remaining energy and momentum.

Background Radiation

Background radiation is always present from natural and artificial sources:

- **Natural sources:** Radon gas from rocks and soil (~50% of total), cosmic rays, rocks and soil, food and drink (e.g. potassium-40 in bananas), internal (carbon-14 and potassium-40 in the body)
- **Artificial sources:** Medical (X-rays, CT scans, nuclear medicine), nuclear weapons testing fallout, nuclear power

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When measuring the activity of a source, you must always subtract the background count rate from your readings to get the corrected count rate.

2. Radioactive Decay and Half-Life

Activity (A): The rate of decay — the number of nuclei decaying per unit time. A = λN. Units: Bq (becquerels). 1 Bq = 1 decay per second. Also: A = −dN/dt.

Decay constant (λ): The probability that a given nucleus will decay per unit time. Units: s^{-1} (or min^{-1}, hr^{-1}, etc.). A large λ means the isotope decays quickly.

Half-life ($t_{½}$): The time taken for the number of radioactive nuclei (or the activity) to decrease to half its original value. It is constant for a given isotope and does not depend on the number of nuclei remaining.

$$\lambda = \ln 2 \,/\, t_{½} = 0.693 \,/\, t_{½}$$

Exponential Decay Equations

$$N = N_0 e^{-\lambda t}$$

$$A = A_0 e^{-\lambda t}$$

$$C = C_0 e^{-\lambda t} \text{ (where C = corrected count rate)}$$

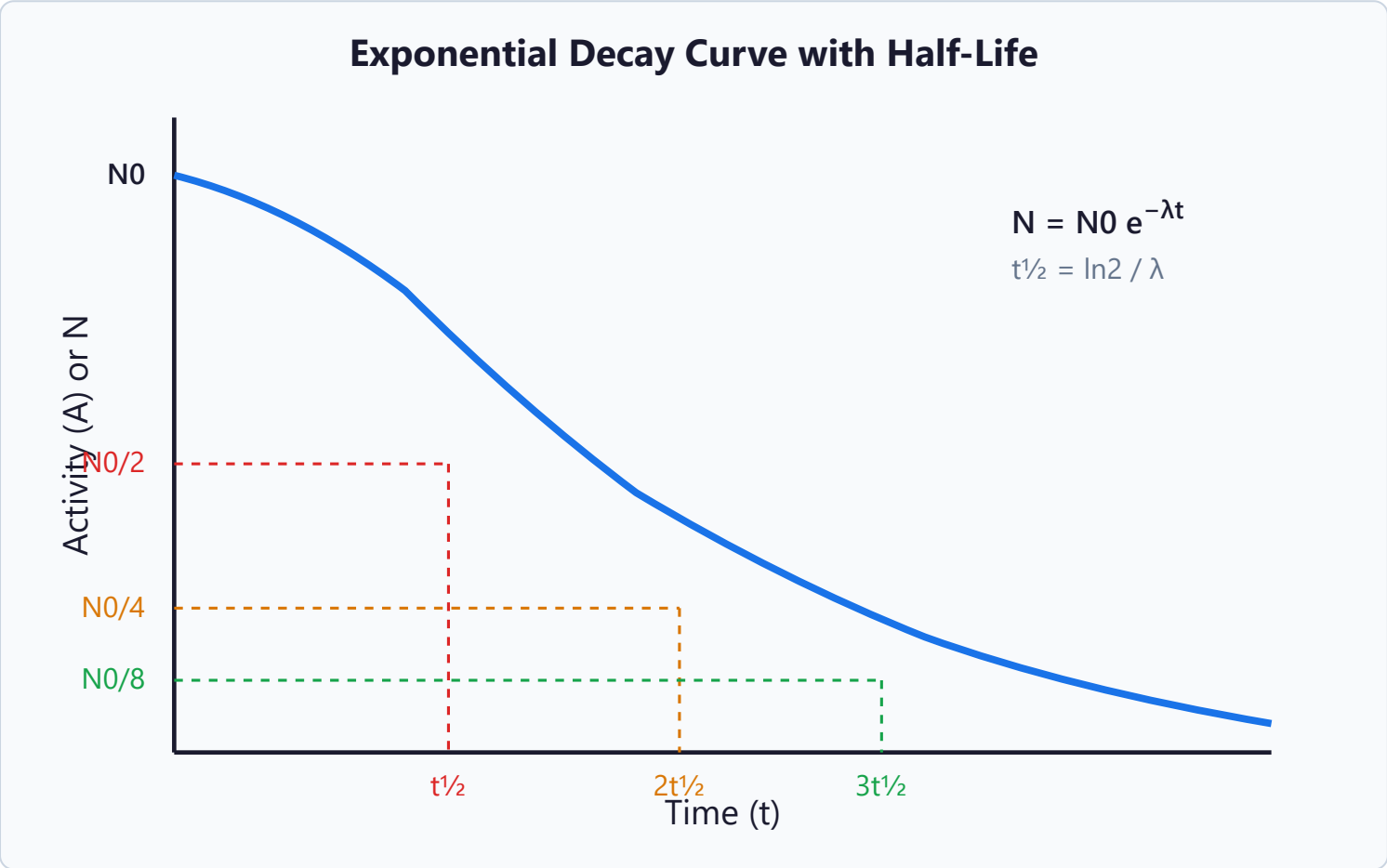
All three quantities (N, A, and corrected count rate) decay with the same exponential law and the same half-life.

Quick Calculation Using Half-Lives

After n half-lives, the fraction remaining is $(1/2)^n$:

Number of half-lives (n)	Fraction remaining	Percentage remaining
1	1/2	50%

2	1/4	25%
3	1/8	12.5%
4	1/16	6.25%
5	1/32	3.125%
10	1/1024	~0.1%



Graphical Methods

- A plot of N (or A) against t gives the characteristic exponential decay curve. Half-life can be read directly from this graph.
- A plot of **ln(N)** against t gives a straight line with gradient = −λ and y-intercept = $\ln(N_0)$. This is the best way to determine λ precisely.
- To verify exponential decay, check that the half-life is constant: measure the time for the value to halve at different starting points on the curve. If it is always the same, the decay is exponential.

Worked Example: A radioactive sample has an initial activity of 640 Bq and a half-life of 8.0 hours. Calculate: (a) the activity after 24 hours, (b) the decay constant, (c) the initial number of radioactive atoms.

Solution:

(a) Number of half-lives = 24/8 = 3. Activity = 640 × $(1/2)^3$ = 640/8 = **80 Bq**

(b) $\lambda = \ln 2 / t_{1/2}$ = 0.693 / (8.0 × 3600) = 0.693 / 28800 = **2.41×10^{-5} s^{-1}**

(c) A = λN, so N = A/λ = 640 / (2.41×10^{-5}) = **2.66×10^7 atoms**

Worked Example (using the exponential formula): A sample has a half-life of 5730 years (carbon-14). A piece of ancient wood has an activity of 0.23 Bq g^{-1}, while living wood has 0.26 Bq g^{-1}. Estimate the age of the wood.

Solution:

$A = A_0 e^{-\lambda t}$, so $A/A_0 = e^{-\lambda t}$

$\ln(A/A_0) = -\lambda t$

$\lambda = \ln 2/5730 = 1.21 \times 10^{-4}$ yr^{-1}

t = −ln(0.23/0.26) / λ = −ln(0.885) / (1.21×10^{-4}) = 0.1222 / (1.21×10^{-4}) = **~1010 years**

Exam Tip: After n half-lives, the fraction remaining is $(1/2)^n$. For quick calculations where the time is a whole number of half-lives, use this method. For non-integer numbers of half-lives, use the exponential formula. Always check your units for λ and t are consistent (both in seconds, or both in years, etc.).

3. Nuclear Radius

Experiments using electron diffraction and alpha-particle scattering show that the nuclear radius depends on the mass number:

$$r = r_0 A^{1/3}$$

- r = nuclear radius (m)

- r_0 = empirical constant ≈ 1.2–1.4 fm (1 fm = 1 femtometre = 10^{-15} m)
- A = mass number (nucleon number)

This relationship tells us that the nuclear volume is proportional to A (since $V = (4/3)\pi r^3 = (4/3)\pi r_0^3 A$), meaning each nucleon occupies approximately the same volume. This implies nucleons are packed together with a roughly fixed separation.

Nuclear Density

Since mass ≈ Au (where u = 1.66 × 10^{-27} kg) and volume = $(4/3)\pi r_0^3 A$:

$$\rho = Au / [(4/3)\pi r_0^3 A] = 3u / (4\pi r_0^3) \approx 10^{17}\ \mathrm{kg\ m^{-3}}$$

The A cancels, meaning **nuclear density is approximately constant for all nuclei**. This is roughly 10^{14} times the density of ordinary matter, confirming that atoms are mostly empty space. Neutron stars have a similar density to nuclear matter.

Electron Diffraction

High-energy electrons (with de Broglie wavelength comparable to nuclear size, ~10^{-15} m) are fired at nuclei. The diffraction pattern shows a central maximum and subsequent minima, from which the nuclear radius can be calculated.

- The first minimum occurs at $\sin\theta \approx 1.22\lambda/(2r)$, allowing r to be determined.
- Electrons are preferred over alpha particles for measuring nuclear radius because they are leptons and do not interact via the strong nuclear force — they only interact electromagnetically, giving a cleaner measurement of the charge distribution.
- For the de Broglie wavelength to be ~10^{-15} m, the electron energy must be very high (~GeV range), requiring relativistic corrections.

Exam Tip: Electron diffraction is used to measure nuclear radius because: (1) electrons are leptons with no strong force interaction, so they probe the charge distribution directly; (2) at high energies their de Broglie wavelength is comparable to nuclear size (~10^{-15} m); (3) the diffraction pattern can be analysed to find the nuclear radius precisely.

4. Mass Defect and Binding Energy

Mass defect (Δm): The difference between the total mass of the individual separated nucleons and the actual mass of the assembled nucleus.

$$\Delta m = [Zm_p + (A - Z)m_n] - m_{nucleus}$$

The mass defect is always positive (the nucleus is lighter than the sum of its parts). The "missing" mass has been converted to binding energy via $E = mc^2$.

Binding energy (BE): The minimum energy required to completely separate all the nucleons in a nucleus to infinity. Equivalently, the energy released when individual nucleons come together from infinity to form the nucleus.

$$BE = \Delta m \times c^2$$

Mass-Energy Equivalence

$$E = mc^2$$

- E = energy (J)
- m = mass (kg)
- c = speed of light = 3.00×10^8 m s^{-1}

Useful Conversions

Quantity	Value
1 atomic mass unit (u)	1.661×10^{-27} kg
1 u in energy	931.5 MeV (via $E = mc^2$)
1 eV	1.60×10^{-19} J
1 MeV	1.60×10^{-13} J

Binding Energy per Nucleon

$$\text{Binding energy per nucleon} = \text{Total binding energy} / A$$

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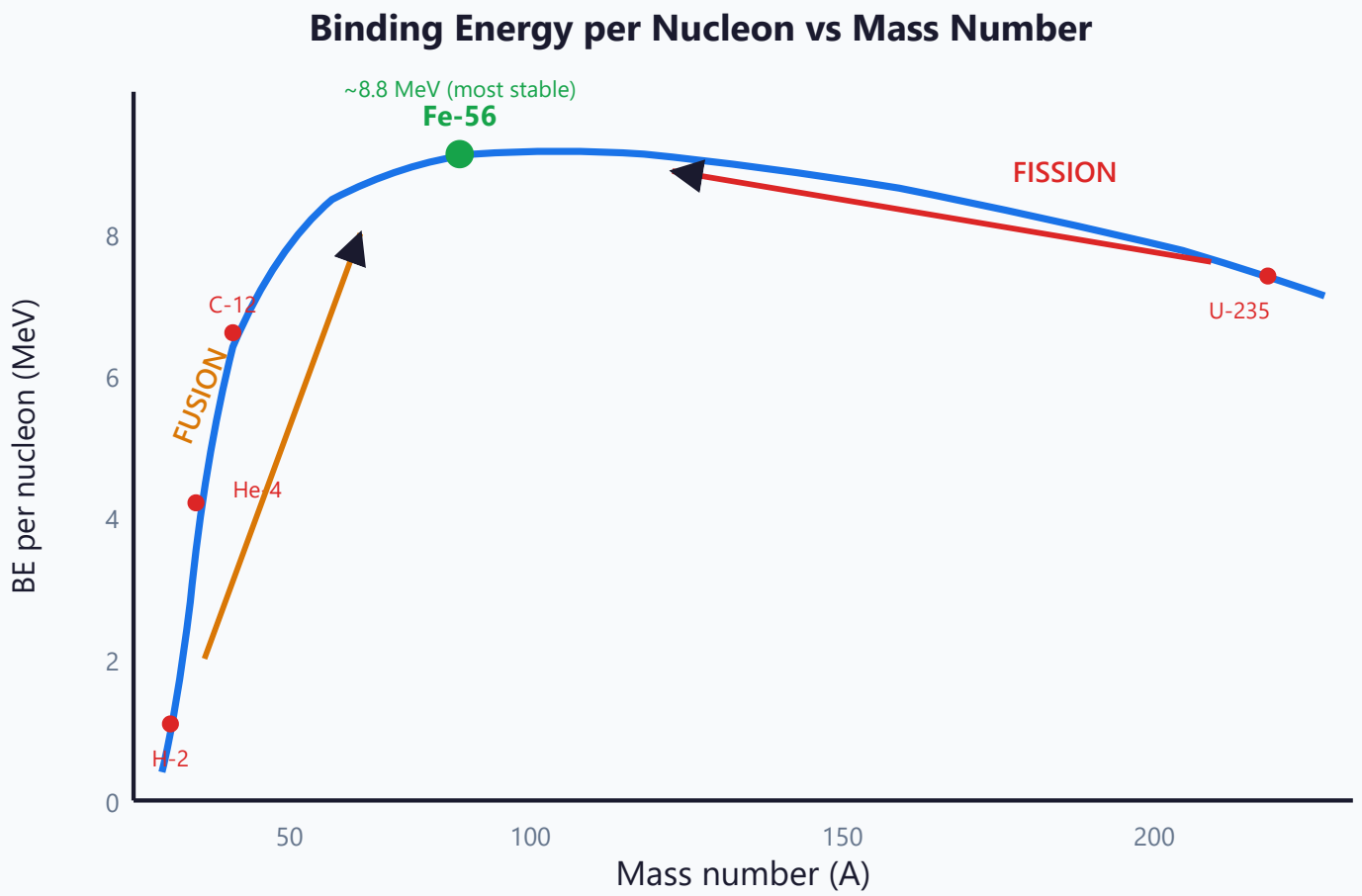
This is the key measure of nuclear stability. Higher binding energy per nucleon = more stable nucleus = more energy needed to remove each nucleon.



Both fusion and fission move nuclei towards Fe-56, releasing energy.

Key Features of the Binding Energy Curve

- The peak is at **iron-56** (and nickel-62), with BE per nucleon ≈ 8.8 MeV. Iron-56 is the most stable nucleus.
- **Light nuclei (A < 56):** BE per nucleon increases with A. These nuclei can release energy by **fusion** (combining to form heavier, more stable nuclei).
- **Heavy nuclei (A > 56):** BE per nucleon decreases with A. These nuclei can release energy by **fission** (splitting into lighter, more stable nuclei).
- Both fusion and fission release energy because the products have higher BE per nucleon than the reactants. The increase in total binding energy equals the energy released.
- He-4 has an unusually high BE per nucleon for its mass, which explains why alpha particles are so stable and commonly emitted.

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Worked Example: Calculate the energy released in the fusion reaction: ${}^{2}_{1}H + {}^{3}_{1}H \rightarrow {}^{4}_{2}He + {}^{1}_{0}n$.

Masses: ${}^{2}H$ = 2.01410 u, ${}^{3}H$ = 3.01605 u, ${}^{4}He$ = 4.00260 u, n = 1.00867 u.

Solution:
Total mass of reactants = 2.01410 + 3.01605 = 5.03015 u
Total mass of products = 4.00260 + 1.00867 = 5.01127 u
Mass defect Δm = 5.03015 − 5.01127 = 0.01888 u
Energy released = 0.01888 × 931.5 = **17.6 MeV**
This is the energy released per fusion event, primarily as kinetic energy of the products.

Exam Tip: Energy released = (total BE of products) − (total BE of reactants). If this is positive, energy is released. Alternatively, energy released = (mass of reactants − mass of products) × c^2 = $\Delta m \times c^2$. Both methods give the same answer. Use whichever data the question provides.

5. Nuclear Fission

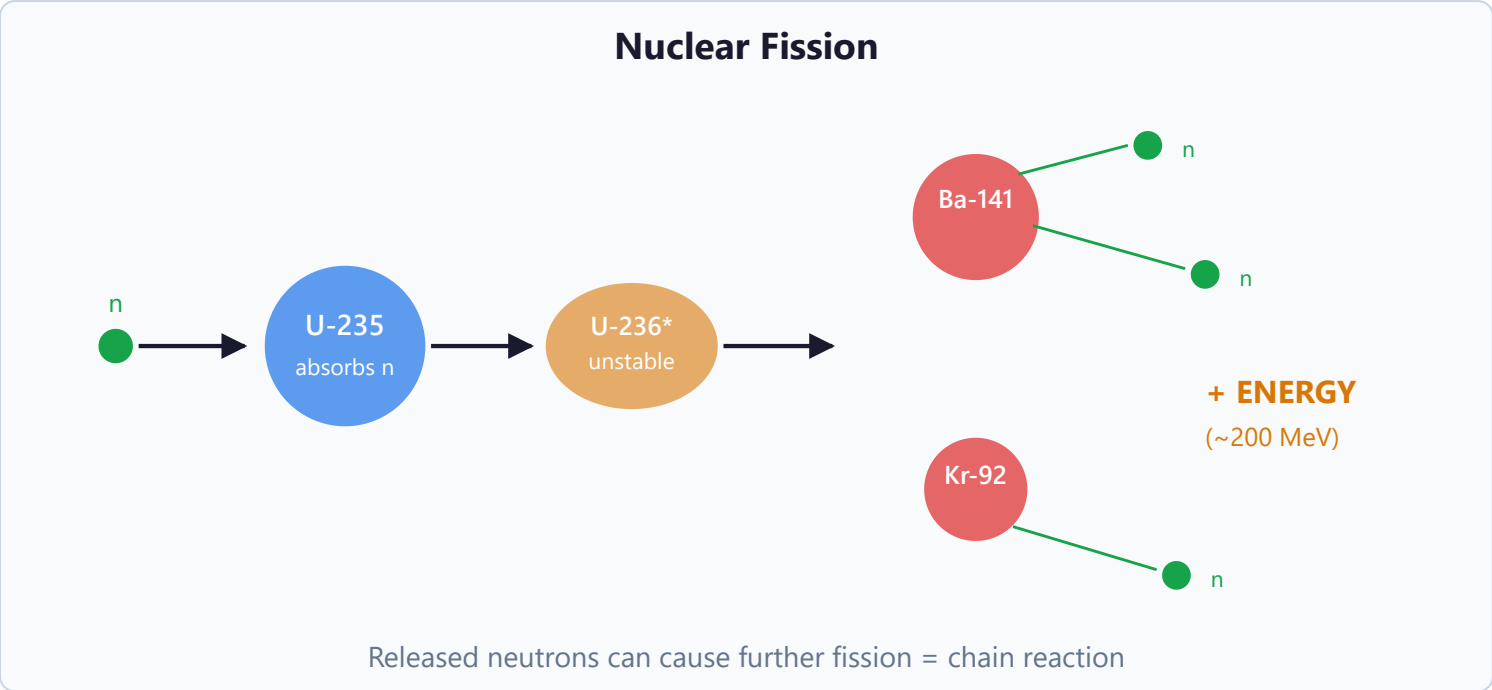
Nuclear fission: The splitting of a heavy nucleus into two smaller daughter nuclei of roughly equal mass, accompanied by the release of neutrons and a large amount of energy (~200 MeV per fission event).

Induced Fission

A thermal (slow) neutron is absorbed by a heavy nucleus (e.g. U-235), forming an unstable compound nucleus (U-236), which then splits. Thermal neutrons are used because they have a much higher probability of being captured than fast neutrons.

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{236}_{92}\mathrm{U} \rightarrow {}^{141}_{56}\mathrm{Ba} + {}^{92}_{36}\mathrm{Kr} + 3\,{}^{1}_{0}\mathrm{n} + \text{energy}$$

(Many different product pairs are possible; this is just one example.)



Chain Reaction

Each fission event releases 2–3 neutrons. These can go on to cause further fission events, creating a chain reaction.

- **Uncontrolled chain reaction:** Every released neutron causes further fission. The number of fissions increases exponentially — a nuclear explosion.
- **Controlled chain reaction:** On average, exactly one neutron per fission goes on to cause another fission. The power output is steady and constant — a nuclear reactor.
- **Critical mass:** The minimum mass of fissile material needed to sustain a chain reaction. Below this mass, too many neutrons escape without causing fission.

Nuclear Reactor Components

Component	Function	Material Examples
Fuel rods	Contain fissile material that undergoes fission	Enriched uranium (UO_2 pellets, ~3–5% U-235) or plutonium
Moderator	Slows down fast neutrons to thermal speeds, increasing the probability of absorption by U-235. Neutrons lose KE through elastic collisions with moderator nuclei.	Water (H_2O), heavy water (D_2O), graphite

Control rods	Absorb excess neutrons to control the rate of the chain reaction. Inserted further to slow the reaction; withdrawn to speed it up.	Boron, cadmium
Coolant	Transfers thermal energy from the reactor core to heat exchangers to generate steam for turbines.	Water, heavy water, CO_2, liquid sodium
Shielding	Absorbs radiation to protect workers and the environment.	Thick concrete and steel

Exam Tip: The moderator must have nuclei of similar mass to neutrons for efficient energy transfer in elastic collisions (like a billiard ball hitting another ball of the same mass). Water contains hydrogen nuclei (protons), which have nearly the same mass as neutrons, making water an effective moderator. Graphite (carbon-12) also works but requires more collisions.

6. Nuclear Fusion

Nuclear fusion: The joining of two light nuclei to form a heavier nucleus, with a release of energy (because the products have higher binding energy per nucleon than the reactants).

Example (deuterium-tritium fusion):

$$^{2}_{1}\mathrm{H} + ^{3}_{1}\mathrm{H} \rightarrow ^{4}_{2}\mathrm{He} + ^{1}_{0}\mathrm{n} + 17.6\ \mathrm{MeV}$$

Conditions for Fusion

- **Extremely high temperature** (~10^7–10^8 K) — needed to give nuclei enough kinetic energy to overcome the electrostatic (Coulomb) repulsion between the positively charged nuclei.
- **High density/pressure** — increases the probability of collisions between nuclei.
- **Sufficient confinement time** — the plasma must be contained long enough for a significant number of fusion reactions to occur.

At these temperatures, matter exists as a **plasma** (a gas of fully ionised atoms — free nuclei and electrons). The main engineering challenge is confining this plasma. Two approaches are being developed:

- **Magnetic confinement:** A tokamak uses powerful magnetic fields to contain the plasma in a toroidal (doughnut) shape (e.g. ITER, JET).
- **Inertial confinement:** Powerful lasers compress a small pellet of fuel to extreme density and temperature.

Fusion in Stars

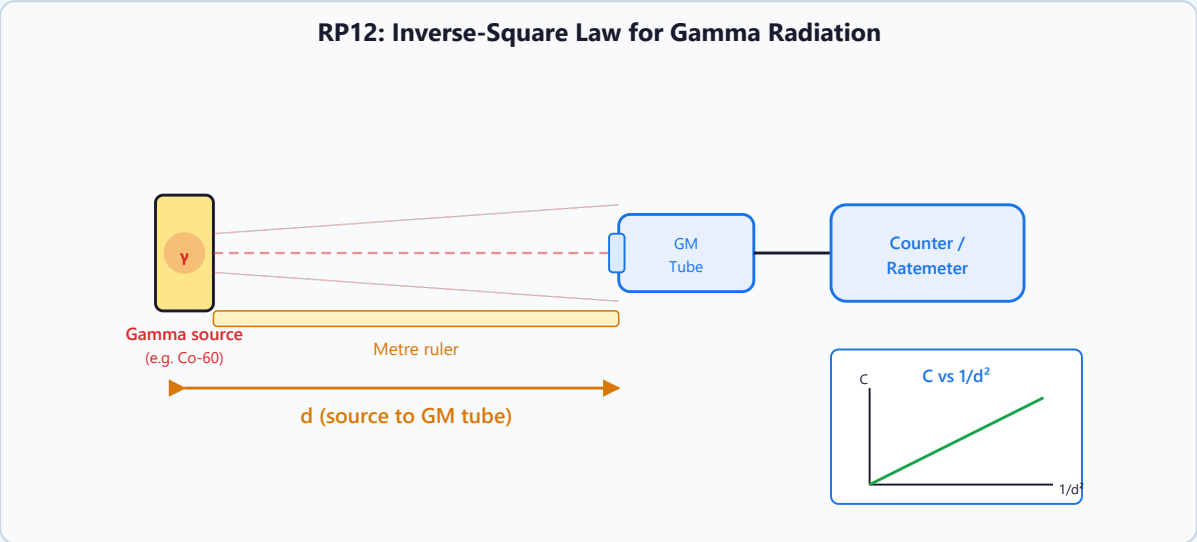
Stars are powered by nuclear fusion. In main-sequence stars like the Sun, hydrogen nuclei fuse to form helium via the **proton-proton chain**. The enormous gravitational pressure in the stellar core provides the necessary temperature and density. The star is in **hydrostatic equilibrium**: the outward radiation pressure from fusion balances the inward gravitational force.

> **Exam Tip:** Fusion releases more energy per kilogram of fuel than fission (because the increase in BE per nucleon is greater for light nuclei fusing). However, achieving the conditions for controlled fusion on Earth remains an enormous engineering challenge. Both fission and fusion release energy because the products have higher binding energy per nucleon than the reactants — both move towards the peak at iron-56.

Summary of Key Equations

Quantity	Equation
Activity	$A = \lambda N = -dN/dt$
Decay constant and half-life	$\lambda = \ln 2 / t_{1/2}$
Exponential decay	$N = N_0 e^{-\lambda t}$; $A = A_0 e^{-\lambda t}$
Nuclear radius	$r = r_0 A^{1/3}$
Mass-energy equivalence	$E = mc^2$; 1 u = 931.5 MeV
Mass defect	$\Delta m = [Zm_p + (A-Z)m_n] - m_{nucleus}$
Binding energy	$BE = \Delta m \times c^2$
Energy released in reaction	$\Delta E = (\text{mass reactants} - \text{mass products}) \times c^2$

⚗ Required Practical 12: Inverse-Square Law for Gamma Radiation



Apparatus

- Sealed gamma source (e.g. cobalt-60 or caesium-137) in a source holder
- Geiger-Muller (GM) tube and counter (or ratemeter)
- Metre ruler or tape measure
- Retort stand and clamp
- Stopwatch (for timed counts)
- Lead absorber set (to verify the source emits gamma)

Method

1. **Background count:** With no source present, measure the background count rate. Run the counter for at least 3 minutes and calculate the mean background count rate C_{bg}.
2. Place the gamma source at one end of the ruler. Place the GM tube at a known distance d from the source, with the tube window facing the source.
3. Count the number of counts N in a fixed time (e.g. 60 s). Calculate count rate $C = N/t$.
4. Subtract background: corrected count rate $= C - C_{bg}$.
5. Move the GM tube further away. Repeat for at least 6–8 different distances (e.g. 5 cm to 50 cm in steps of 5 cm).

6. Plot corrected count rate vs $1/d^2$. A straight line through the origin confirms the inverse-square law.
7. Alternatively, plot log(C) vs log(d). The gradient should be −2.

Safety

- Handle the source with long tongs — never with bare hands.
- Keep the source in a lead-lined container when not in use.
- Point the source away from people. Maximise distance and minimise exposure time.
- Only a teacher or authorised person should handle the source.
- Wash hands after the experiment. Record the source in the radioactive materials register.

Analysis

For a point source emitting gamma uniformly: I (and hence count rate C) follows the inverse-square law:

$$C \propto 1/d^2 \;\Rightarrow\; C = k/d^2$$

Plotting corrected C vs $1/d^2$ gives a straight line through the origin. If the line does not pass through the origin, the effective centre of the source may not be at the measured position. Introduce a correction d_0: $C = k/(d + d_0)^2$.

Sources of Error & Improvements

Source of Error	Type	Improvement
Count rates fluctuate (radioactive decay is random)	Random	Use long counting times (≥ 60 s); repeat and average
Background radiation not properly subtracted	Systematic	Measure background carefully (3+ minutes); subtract from each reading
Distance d measured to wrong point on GM tube	Systematic	Measure to the GM tube window; apply correction factor d_0
Source is not a true point source	Systematic	Use a small sealed source; work at distances much larger than source diameter

Dead time of GM tube (misses counts at high rates)	Systematic	Avoid very close distances; apply dead-time correction if known

Exam Tip: Always subtract background count rate before plotting. The randomness of radioactive decay means count rates fluctuate — this is why you take long counting times and repeat measurements. If the graph of C vs $1/d^2$ does not pass through the origin, this indicates a systematic error in d. You can find d_0 by plotting $\sqrt{(1/C)}$ vs d, which should be linear with an x-intercept of $-d_0$.